

## Ewerton Rocha Vieira

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### **SUMMARY**

Senior research scientist with deep applied-mathematics expertise in automation, robot control, sensing, sensor fusion, and embedded pipelines.

- Strong expertise in flow matching and diffusion-based methods for robotic applications.
- Robotic Arm Imitation Learning Using Wearable Human Motion Data.
- Hands-on experience with UR5e and Baxter robotic platforms, developing manipulation tasks and sim-to-real pipelines for transferring control policies and robotic behaviors from simulation to physical systems.
- Decentralized Multi-Agent Task Allocation for Autonomous Underwater Vehicle Swarms.
- Proven record of success in transdisciplinary collaborations across academia and industry.
- Skilled in developing algorithms to evaluate robot-control performance, robustness, and mathematical models.
- Architected advanced sensing, fusion, and embedded-system processing pipelines.
- Conducted mathematical and statistical data analysis and data modeling using Gaussian processes.
- Excellent oral and written communicator; 5 first-author robotics conference papers, 5 mathematical journal papers, and 1 submission.
- Presented research at 17 international conferences.

**Technical Skills:** Python, C++, MATLAB, Linux, Git, Docker; PyTorch, scikit-learn, GPy; ROS/ROS 2, MoveIt; Gazebo and MuJoCo; robot manipulation, motion planning, inverse kinematics, control, reinforcement learning, imitation learning, domain randomization, and sim-to-real transfer.

**Open Projects:** [github.com/Ewerton-Vieira/MORALS](https://github.com/Ewerton-Vieira/MORALS), [github.com/Ewerton-Vieira/flow\\_matching](https://github.com/Ewerton-Vieira/flow_matching), [github.com/PRX-Kinodynamic/Generative-Motion-Planner](https://github.com/PRX-Kinodynamic/Generative-Motion-Planner), [github.com/PRX-Kinodynamic/ML4KP-devel](https://github.com/PRX-Kinodynamic/ML4KP-devel), [github.com/DanManN/planning\\_baxter](https://github.com/DanManN/planning_baxter).

### **EDUCATION**

*Doctor of Philosophy*, Mathematics, Federal University of Campinas, Brazil, SP | 2015

*Master's*, Mathematics, Federal University of Campinas, Brazil, SP | 2011

*Bachelor of Mathematics*, University of Brasília, Brazil, SP | 2008

### **PROFESSIONAL EXPERIENCE**

**Research and Development Senior Engineer, Thinksense, Arlington, VA | July 2025 – Present**

- Collected and processed IMU-based human motion data to create demonstration trajectories for robotic-arm imitation learning.
- Developed diffusion and flow-matching policies to translate smartphone and smartwatch IMU data into smooth robotic motions that replicate human movements.
- Led mSensor, an Android app for context-aware sensor fusion and seamless sensor integration.
- Architected end-to-end advanced sensing, fusion, and embedded processing pipelines for real-time systems.
- Specialized in biomedical signal processing and wearable systems for robust, real-world monitoring.
- Developed low-power multi-modal AI monitoring on wearables and smartphones, optimizing on-device inference.
- Designed path planning and distributed autonomy for swarms of unmanned underwater vehicles (UUVs).
- Led the scientific team for ENCASE (Encapsulation of Naval Combatants through Autonomous Swarm Engagement), advancing autonomous swarm engagement concepts and implementation.
- Built underwater simulation, navigation, and mission-rehearsal frameworks for autonomy development and testing.
- Led the scientific team for COMBAT (Continuous Multi-Factor Authentication using Biosignals and Gait for SWaP Defense Equipment), delivering continuous authentication for resource-constrained lightweight IoT devices.

**Post-Doctoral Associate, DIMACS (the Center for Discrete Mathematics and Theoretical Computer Science) and Department of Mathematics, Rutgers University, NJ | September 2020 – June 2025**

Supervisors: Prof. Konstantin Mischaikow (Mathematician), Prof. Kostas Bekris (Robotist), Prof. Ying Hung (Statistician).

- Evaluated the robustness and performance of black-box systems, such as robot controllers with policies learned through reinforcement learning.
- Computed safe operational regions for deploying robot controllers.
- Synthesized hybrid controllers capable of operating over a wider range of the state space.
- Designed high-level and robust solutions for robot task manipulations such as object retrieval.

- Created surrogate models (data-driven and statistical methods) based on data collected from robot control.
- Optimized data volume required for system evaluation by integrating statistical modeling and optimization methods.
- Developed adaptive schemes for collecting data to minimize uncertainty in model systems.
- Taught Applied Math at Rutgers focused on Data-Driven Science and Engineering: Machine Learning, Dynamical Systems, and Control.

**Assistant Professor of Mathematics, Federal University of Goiás, Brazil, GO | March 2015 – August 2020**

- Taught Calculus, Differential Equations, Statistics, Probability, Applied Statistics for Finance, Topology, Dynamical Systems, and Algebraic Topology.
- Lead workshops on recent advances in mathematics.
- Supervised students and guided their dissertations.
- Collaborated and coordinated with inter-departmental team to promote research and development.
- Assisted in management of the Institute of Mathematics and Statistics.

**PUBLICATIONS**

***Peer Reviewed Manuscripts Submitted***

1. D. Ramesh, S. Tangirala, M. Ying, **E. R. Vieira**, Y. Hung and K. E. Bekris. *Data-Efficient RoA Verification of Black-box Controllers via Flow Matching and Conformal Inference*.

***Peer Reviewed Manuscripts***

1. **E. R. Vieira**, A. Sivaramakrishnan, S. Tangirala, E. Granados, K. Mischaikow, K. E. Bekris, *MORALS: Analysis of High-Dimensional Robot Controllers via Topological Tools in a Latent Space*, 2024 IEEE International Conference on Robotics and Automation (ICRA). (**Best Paper Award in Automation finalist**).
2. **E. R. Vieira**, A. Sivaramakrishnan, Y. Song, E. Granados, M. Gameiro, K. Mischaikow, Y. Hung and K. E. Bekris. *Data-Efficient Characterization of the Global Dynamics of Robot Controllers with Confidence Guarantees*. IEEE International Conference on Robotics and Automation (ICRA).
3. **E. R. Vieira**, K. Gao, D. Nakhimovich, K. E. Bekris and J. Yu. *Persistent Homology Guided Monte-Carlo Three Search for Effective Non-Prehensile Manipulation*. ISER (International Symposium on Experimental Robotics)-Chiang Mai, Thailand, 2023.
4. B. Batko, M. Gameiro, Y. Hung, W. Kalies, K. Mischaikow and **E. R. Vieira**, *Identifying Nonlinear Dynamics with High Confidence from Sparse Data*, SIAM Journal on Applied Dynamical Systems 23 (1), 383-409.
5. **E. R. Vieira**, E. Granados, A. Sivaramakrishnan, M. Gameiro, K. Mischaikow and K. E. Bekris. *Morse Graphs: Topological Tools for Analyzing the Global Dynamics of Robot Controllers*. Workshop on the Algorithmic Foundations of Robotics 2022, WAFR-22.
6. **E. R. Vieira**, D. Nakhimovich, K. Gao, R. Wang, J. Yu and K. E. Bekris. *Persistent Homology for Effective Non-Prehensile Manipulation*. International Conference on Robotics and Automation (ICRA) 2022, pp. 1918–1924.
7. D. Lima, M. Silveira, **E. R. Vieira**. *Covering action on Conley index theory*. Ergodic Theory and Dynamical Systems, 1-33, 2022. doi:10.1017/etds.2022.13
8. R. Bastos, N.R. Rocco and **E. R. Vieira**. *Finiteness of homotopy groups related to the non-abelian tensor product* Annali di Matematica (2019) No 1, 1--11.
9. R. Franzosa, **E. R. Vieira**. *Transition Matrix Theory*, Transactions of the American Mathematical Society, 369 (2017) 11, 7737–7764.
10. R. Franzosa, K. A. de Rezende, **E. R. Vieira**. *Generalized topological transition matrix*. Topological Methods in Nonlinear Analysis 48 No 1 (2016), 183--212.

**SELECTED CONFERENCE TALKS**

1. International Conference on Robotics and Automation (ICRA) 2022 & 2024.
2. International Symposium on Experimental Robotics (ISER) - Chiang Mai, Thailand, 2023.
3. International Congress on Industrial and Applied Mathematics ICIAM 2023 Tokyo.
4. Workshop on the Algorithmic Foundations of Robotics (WAFR) 2022. Morse Graphs: Topological Tools for Analyzing the Global Dynamics of Robot Controllers. University of Maryland, College Park, USA.
5. Workshop on Dynamics, Topology, and Robotic Control. Learning Global Dynamics from Data. 2021

**STUDENTS SUPERVISED**

Supervised eleven undergraduate students, ten graduate students (nine master's students and one co-advisor Ph.D student).